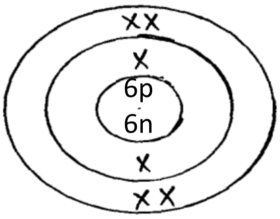
233/1
CHEMISTRY PAPER 1
MARKING SCHEME

1. (a) isotopes √1 mark

   (b)



   (c) XO_2
   √½

2. Add excess Lead (II) carbonate √½ mark to dilute nitric acid √½ mark then filter off the unreacted Lead (II) carbonate. Obtain the filtrate which is Lead (II) nitrate. Add Sodium Chloride √½ mark solution to the filtrate. A white precipitate forms which is Lead (II) chloride. Filter the precipitate and wash it with distilled water and dry it between filter paper. (3mks)

3. Molarity of $Na_2CO_3 = \frac{5.3}{106} = 0.5 moles/dm^3$

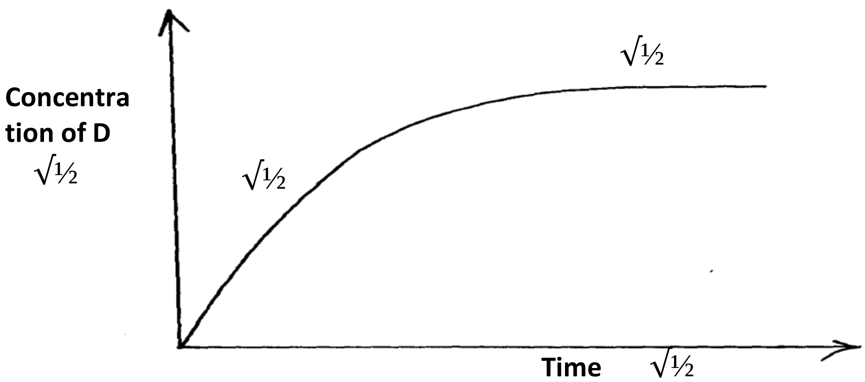
$$Na_2CO_{3(aq)} + 2HCL_{(aq)} \rightarrow 2Nacl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$$

$$\frac{M_aV_a}{M_bV_b} = \frac{2}{1}$$

$$\frac{0.5 \times Va}{0.05 \times 20} = 2$$

$$Va = \frac{2 \times 0.05 \times 20}{0.5} = 4cm^3$$

4. (a)



   (b) Concentration of gas D increases from time(√½ mark) zero as C is converted to D until equilibrium is (√½ mark) reached after which the concentration of D remains constant.

5. – Leaves no wetness

- It is a better coolant

6. Molecules of water = $\dfrac{20.3-9.5g}{18} = 0.6moles$

Mole ratio Y:N_2O

1:6

X:0.6

Molar mass of Y = $\dfrac{9.5}{0.1} = 95$

R.F.M of hydrated salt = $95+18\times6$

95+108 = 203

7. (a) To prevent sacking back of Lead (II) nitrate solution. √1mark

(b) A white precipitate forms.(√1mark) HCL gas dissolves in the solution then reacts with Lead (II) nitrate to form insoluble Lead (II) chloride.

8. Element C. It has small atomic radius hence strong nuclear attraction on electrons.

9. (a)
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H   H
|   |
C = C
|   |
H   Cl
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(b) Molar mass of monometer = 24.5+3+35.5 = 62.5

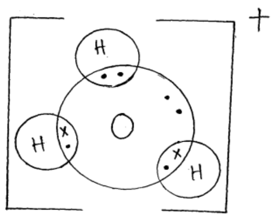
No. of monometer molecules = $\dfrac{28125}{62.5} = 450$

10. Dissolve the mixture in distilled water at room temperature. Heat the solution to 60°C. V will remain in solution while U will crystallise out. Filter out U and dry. Evaporate the filtrate to obtain crystals of V.

11. (a) (i) Cu^{2+} ions

(ii) Fe^{3+} ions

12. (a)



(b) Molecules are held together by weak van der waal forces of attraction.

13. . Al_2O_3

(b) Mgo has a higher m.p than P_4O_{10} MgO has giant ionic structure with strong ionic bond while P_4O_{10}has simple molecular structure with weak intermolecular bonds.

14.

$${}^{32}_{16}S + {}^{1}_{0}n \rightarrow {}^{32}_{15}D + {}^{1}_{1}P \ \surd\text{1mark}$$

(b) 20g $\xrightarrow{\text{12.8 days}}$ 10g $\xrightarrow{\text{12.8 days}}$ 5g $\xrightarrow{\text{12.8 days}}$ 2.5g $\xrightarrow{\text{12.8 days}}$ 1.2 g

Mass remaining = 1.25g √1mark

15. (a) To produce ammonia fumes without decomposing it.

(b) Blue flame

(c) $4NH_{3(g)} A_3O_{2(g)} \rightarrow 2N_{2(g)} + 6H_2O_{(g)}$ √1mark

16 (a) $Q_{(s)} + R^{2+}_{(aq)} \rightarrow Q^{2+}_{(aq)} + R_{(s)}$ √1mark

(b) $+0.34 - (-0.76) = +1.10V$ √1mark

17. Y is a strong acid while X is a weak acid.
Y ionizes fully producing more ions that conduct electricity.

18. (a) Heat √1

(b) (i) Sulphur(IV)oxide √1

(ii) In dry cells √1

Galvanized iron sheets √1

19. $CaCO_{3(s)} \xrightarrow{\text{heat}} CaO_{(s)} + CO_{2(g)}$

$\frac{2.5}{100} = 0.025$

Mole of CO_2 produced = 0.025 (1:1)

Volume of CO_2 = 0.025 x 22.400

= $600cm^3$

20. (i) Aluminium sulphate

(ii) Barium sulphate

(b) $Al^{3+}_{(aq)} + 3OH^-_{(aq)} \longrightarrow Al(OH)_{3(s)}$

21. (a) Sublimation

(b) Oxidation

(c)Dehydration

22. $C_{(s)}$ + $2H_{2(g)}$ ΔH (CH) $CH_{4(g)}$

$\Delta Hc = -3933$ $\Delta Hc = -285kjm$ $\Delta Hc = -90Kjmol$

O2 $+O_2$ $2O_2$

$CO_{2(g)} + 2H_2O$

$\Delta H_7^{\theta}(CH_4) = [\,-393.4 + (-285 \times 2)] - (-890)$

$-963.5 + 890$
$= 73.5\ Kjmol^{-1}$

23. (a) Sample II; it lathers readily with a little soap.
(b) The water Has temporary hardness which is removed by boiling. Hence soap is used after boiling.

24. $Rso_2 = 15cm^3/s$ $Ro_2 = \frac{100}{t}$ cm3/s

$Mso_2 = 64$ $Mo_2 = 32$

$$\frac{Ro_2}{Rso_2} = \sqrt{\frac{Mso_2}{Mo_2}}$$

$$\frac{\frac{100}{t}}{15} = \sqrt{\frac{64}{32}}$$

$$\frac{100}{t} = 15x\sqrt{\frac{64}{32}}$$

$$\frac{100}{t} = 21.2132$$

$$\Rightarrow t\,\frac{100}{21.2132} = 4.7140\,\mathrm{sec}onds$$

25. (a) Calcium hydroxide

(b) $Ca(OH)_{2\,(aq)} + Cl_{2(g)} \longrightarrow CaOCl_{2(s)} + H_2O_{\,(l)}$